Two-Equation Turbulence Models and Effect of Compressibility

Problem

In this homework, the simulation of a turbulent mixing layer is performed using three different turbulence models: the $k-\varepsilon$, $k-\omega$ standard and SST models. The goal is to assess the performance of these turbulence models by comparing their predictions against experimental measurement data and theoretical expectations. The experimental data, computational grid, and boundary conditions necessary for the simulation are provided at: `https://turbmodels.larc.nasa.gov/delvilleshear_val.html`.

The specific objectives of the homework are:

- To simulate the turbulent mixing layer using the $k-\varepsilon$, $k-\omega$ and $k-\omega$ SST turbulence models.
- To compare the computed CFD results with theoretical and experimental results for:
  - Mean velocity profiles at streamwise locations x = 1 mm, x = 200 mm, and x = 650 mm.
  - Reynolds stress profiles at streamwise locations x = 200 mm and x = 650 mm.

Solution

The simulation of turbulent mixing layers plays a crucial role in understanding the fundamental behavior of shear-driven flows, which are common in many engineering applications such as jet propulsion, combustion systems, and atmospheric boundary layers. A mixing layer forms when two parallel streams of fluid with different velocities interact, leading to the development of a shear region characterized by strong velocity gradients and turbulent fluctuations.

The objective of this study is to evaluate and compare the ability of various turbulence models to accurately capture the key features of a turbulent mixing layer, particularly the growth of the shear layer and the distribution of turbulent stresses.

Setup

The computational study was performed using ANSYS Fluent in a 2D mixing layer validation case. The computational domain represents a two-dimensional planar mixing layer, where two streams with slightly different total pressures and temperatures are introduced into the domain through pressure boundary conditions, presented on the Fig. 1. The simulation employed a steady-state, pressure-based solver.

Figure 1 shows the boundary conditions applied in the domain:

- **Inlet:** The top inflow condition was applied with $P/P_{ref} = 1.01031$ and $T_t/T_{ref} = 1.00293$ at $x = -500$ mm, with meaning that initial supersonic and total gauge pressure were 101031 Pa at the top inlet and temperature 293.86 K. The bottom inflow condition was applied with $P/P_{ref} = 1.00299$ and $T_t/T_{ref} = 1.00085$ at $x = -300$ mm, with meaning that initial supersonic and total gauge pressure were 100299 Pa at the bottom inlet and temperature 293.25 K. These reference conditions are consistent with the 2D Mixing Layer Experiment.
- **Top and Bottom Boundaries:** Slip wall conditions are applied at the top and bottom boundaries, allowing tangential flow while preventing normal flow. No shear stress is imposed (adiabatic slip walls).

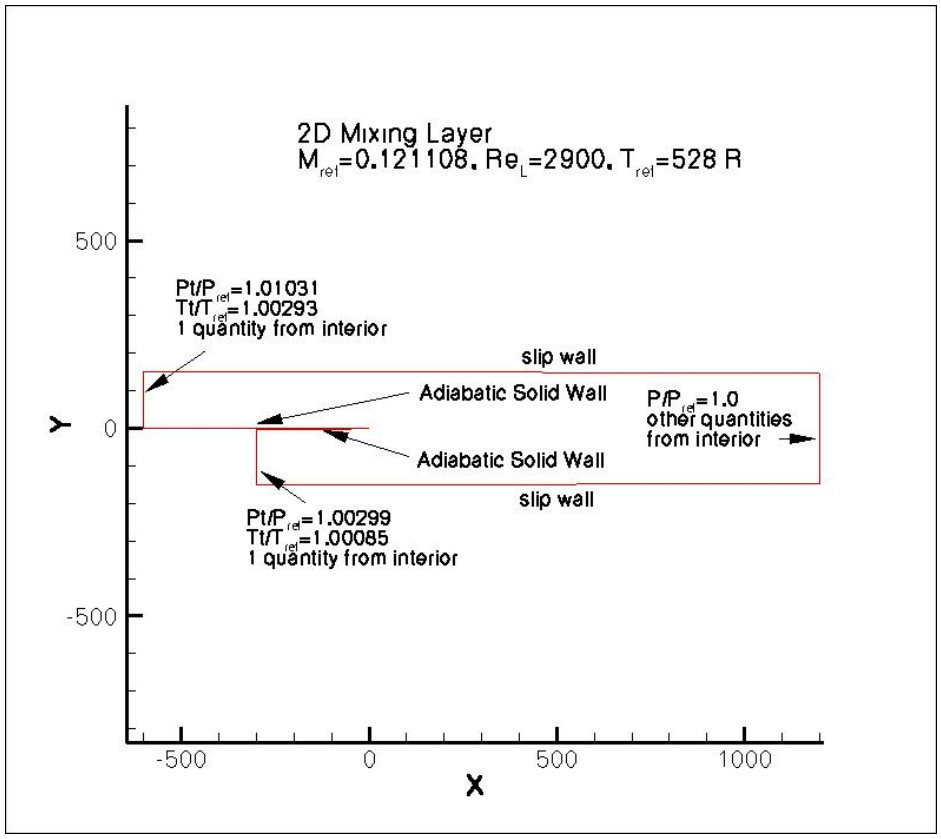


Figure 1: Computational domain and boundary conditions for the mixing boundary layer simulation case, taken from [1].

- **Walls:** The solid walls adjacent (red lines in the Fig. 1)to the shear layer are defined as adiabatic solid walls with no normal velocity (impermeable) but allowing tangential flow (slip condition).
- **Symmetry:** The domain's lower boundary, along $y = 0$, was modeled as the axis of symmetry.
- **Outlet:** At the outlet, a pressure outlet condition was applied with $P/P_{ref} = 1.0$.

Turbulence is modeled using three two-equation models:$k-\varepsilon$, $k-\omega$, and $k-\omega$ SST (Shear Stress Transport) models. Second-order discretizations schemes are used for spatial derivatives to minimize numerical diffusion. Turbulence quantities at the inlet are initialized based on estimated turbulence intensity and length scales corresponding to the mixing layer conditions.

Postprocessing was performed in Ansys Fluent and Python to extract relevant flow quantities along z-axis profiles at several x-axis locations from the inlet. The primary objective was to evaluate the ability of various turbulence models to accurately capture the key features of a turbulent mixing layer.

The following non-dimensional quantities were extracted and plotted at $x = 1$, 200 and 650 mm:

- Streamwise velocity: u
- Reynolds stresses: $\overline{u'u'}, \overline{u'v'}, \overline{v'v'}$.

The Reynolds shear stresses were calculated using custom-defined field functions in Fluent as follows:

$$\overline{u'u'} = \frac{-\mu_t}{\rho}\left(2\frac{\partial u}{\partial x}\right) + \frac{2}{3}k \tag{1}$$

$$\overline{u'v'} = \frac{-\mu_t}{\rho}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) \tag{2}$$

$$\overline{v'v'} = \frac{-\mu_t}{\rho}\left(2\frac{\partial v}{\partial y}\right) + \frac{2}{3}k \tag{3}$$

$$\tag{4}$$

The results from the CFD simulations were plotted alongside experimental data provided in a structured dataset, allowing direct comparison at each downstream location.

Standard $k - \varepsilon$

Two-equation turbulence models allow the determination of both, a turbulent length and time scale by solving two separate transport equations. It is a semi-empirical model, and the derivation of the model equations relies on phenomenological considerations and empiricism. The standard $k - \varepsilon$ model is a model based on model transport equations for the turbulence kinetic energy k and its dissipation rate ε. The model transport equation for is derived from the exact equation, while the model transport equation for was obtained using physical reasoning and bears little resemblance to its mathematically exact counterpart.

In the derivation of the $k - \varepsilon$ model, the assumption is that the flow is fully turbulent, and the effects of molecular viscosity are negligible. This model is therefore valid only for fully turbulent flows. The turbulence kinetic energy, k, and its rate of dissipation, ε, are obtained from the following transport equations [2]:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] + G_k + G_b - \rho\varepsilon - Y_M + S_k$$

$$\frac{\partial}{\partial t}(\rho \varepsilon) + \frac{\partial}{\partial x_i}(\rho \varepsilon u_i) = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right] + C_{1\varepsilon}\frac{\varepsilon}{k}(G_k + C_{3\varepsilon}G_b) - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k} + S_\varepsilon$$

In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. $C_{1\varepsilon}$, $C_{2\varepsilon}$, and $C_{3\varepsilon}$ are constants. σ_k and σ_ε are the turbulent Prandtl numbers for k and ε, respectively. and are user-defined source terms.

$C_{1\varepsilon}$=1.44, $C_{2\varepsilon}$=1.92, and C_μ=0.09, σ_k=1.0 and σ_ε=1.3 default values have been determined from experiments for fundamental turbulent flows including frequently encountered shear flows like boundary layers, mixing layers and jets as well as for decaying isotropic grid turbulence. They have been found to work fairly well for a wide range of wall-bounded and free shear flows.

The turbulent viscosity, μ_t, is computed by combining as follows:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \tag{5}$$

Standard $k-\omega$

The standard $k-\omega$ model in Ansys Fluent is based on a $k-\omega$ model proposed by Wilcox, which incorporates modifications for low-Reynolds number effects, compressibility, and shear flow spreading. One of the weak points of the 1998 Wilcox model is the sensitivity of the solutions to values for k and ω outside the shear layer (freestream sensitivity), which can have a significant effect on the solution, especially for free shear flows [3]. The turbulence kinetic energy, k, and the specific dissipation rate, ε, are obtained from the following transport equations:

$$\frac{\partial}{\partial t}(\rho k)+\frac{\partial}{\partial x_i}(\rho k u_i)=\frac{\partial}{\partial x_j}\left(\Gamma_k \frac{\partial k}{\partial x_j}\right)+G_k-Y_k+S_k+G_b$$

$$\frac{\partial}{\partial t}(\rho \omega)+\frac{\partial}{\partial x_i}(\rho \omega u_i)=\frac{\partial}{\partial x_j}\left(\Gamma_\omega \frac{\partial \omega}{\partial x_j}\right)+G_\omega-Y_\omega+S_\omega+G_{\omega b}$$

In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_ω represents the generation of ω. Γ_k and Γ_ω represent effective diffusivity of k and ω, respectively. Y_k and Y_ω represent the dissipation of k and ω due to the turbulence. S_k and S_ω are user-defined source terms. G_b and $G_{\omega b}$ account for buoyancy.

The turbulent viscosity, μ_t, is computed by combining as follows:

$$\mu_t = \alpha^* \frac{\rho k}{\omega}, \qquad \alpha^* = 1.0. \tag{6}$$

Shear-Stress Transport (SST) $k-\omega$

The SST $k-\omega$ model accounts for the transport of the turbulence shear stress in the definition of the turbulent viscosity. These features make the SST model more accurate and reliable for a wider class of flows (for example, adverse pressure gradient flows, airfoils, transonic shock waves) than the above mentioned standard models [4]. Other models fail to properly predict the onset and amount of flow separation from smooth surfaces. The main reason is that other models do not account for the transport of the turbulent shear stress. This results in an overprediction of the eddy-viscosity. The proper transport behavior can be obtained by a limiter to the formulation of the eddy-viscosity:

$$\mu_t = \frac{\rho k}{\omega} \frac{1}{max[\frac{1}{\alpha^*}, \frac{SF_2}{a_1 \omega}]}, \tag{7}$$

where S is the strain rate magnitude and α^* is defined as

$$\alpha^* = \alpha^*_\infty \left(\frac{\alpha^*_0 + Re_t/R_k}{1 + Re_t/R_k}\right), \quad Re_t = \frac{\rho k}{\mu \omega}, \quad R_k = 6, \quad \alpha^*_0 = \frac{0.072}{3}. \tag{8}$$

F_2 is given by

$$F_2 = \tanh \Phi_2^2, \qquad \Phi_2 = max\left[2\frac{\sqrt{k}}{0.09\omega y}, \frac{500\mu}{\rho y^2 \omega}\right], \tag{9}$$

where y is the distance to the next surface.

Results & Discussion

Figure 2 shows the velocity magnitude distribution across the 2D mixing layer domain. Two distinct freestream regions are clearly visible:

- The upper stream with a higher velocity (around 42 m/s),
- The lower stream with a lower velocity (around 17 m/s).

At the interface between these two streams, a mixing layer develops, characterized by a gradual transition of velocity values between the two streams. The thickness of this shear region increases as the flow progresses downstream, illustrating the growth of the mixing layer due to turbulent diffusion. The overall distribution is smooth and symmetric, consistent with expectations for a mixing layer under nearly incompressible flow conditions (low Mach number). No visible numerical instabilities or nonphysical oscillations are observed, indicating that the simulation setup and boundary conditions were properly defined.

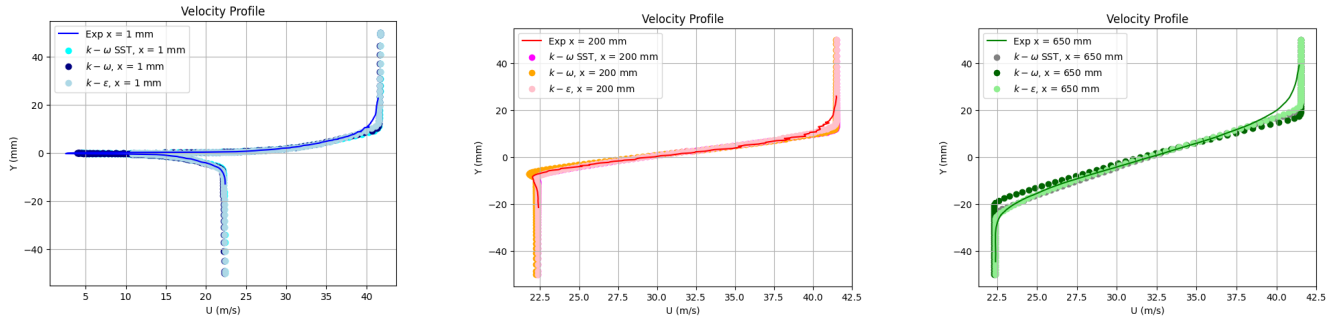
Figure 2: Velocity magnitude distribution over domain.

Velocity profile at different streamwise locations

In the simulation of the mixing layer, velocity profiles were extracted at three streamwise locations: $x = 1$ mm, 200 mm, and 650 mm. The profiles show significant evolution as the flow develops downstream as depicted on Fig. 3.

At x = 1 mm, Fig. 3a, very close to the inlet, the two streams have not yet mixed significantly. The velocity profile exhibits a sharp gradient, closely resembling a step change between the two free-stream velocities, indicating a very thin shear layer with minimal turbulent mixing. For x = 200 mm, Fig. 3b, the shear layer has grown thicker due to the effects of turbulent mixing. The velocity transition between the two streams becomes smoother, and the flow starts to exhibit typical turbulent mixing behavior, with a broader region of intermediate velocities.

Considering x = 650 mm, the shear layer has thickened even further, and the velocity profile becomes more self-similar, meaning its shape, when properly normalized, does not change significantly with further downstream distance. The centerline region shows a much smoother gradient, indicating a high degree of mixing between the two streams. The differences between the profiles at these three locations are primarily due to the growth of the mixing layer caused by turbulent momentum transfer, which spreads the velocity differences over an increasingly wider region as the flow progresses downstream.



(a) Velocity profile at $x = 1$ mm streamwise location.

(b) Velocity profile at $x = 200$ mm streamwise location.

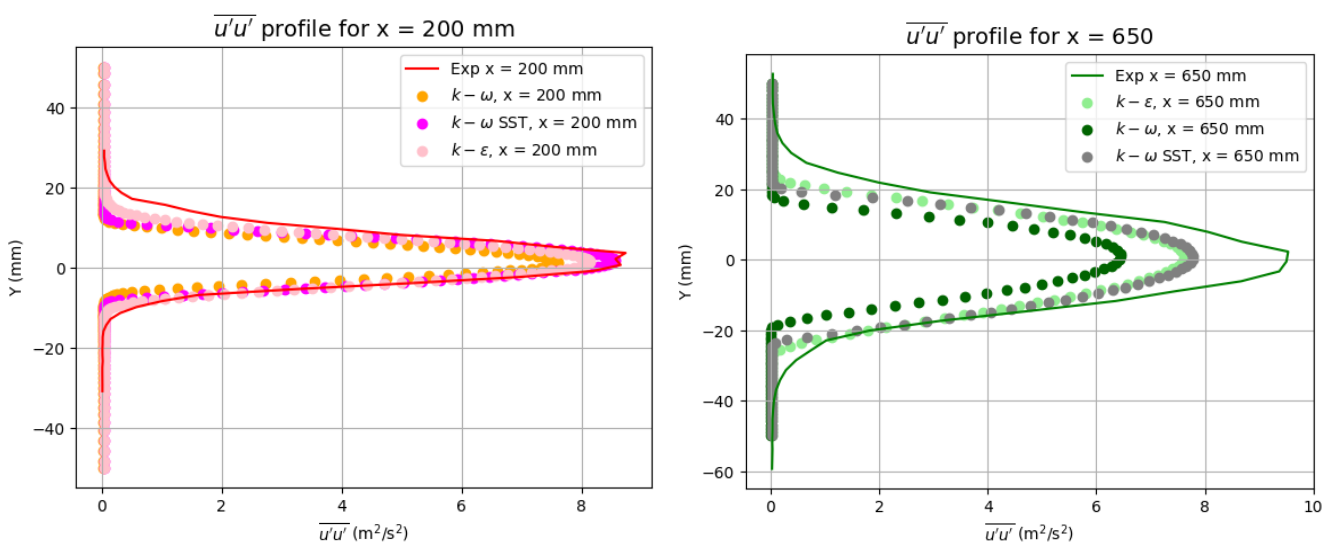
(c) Velocity profile at $x = 650$ mm streamwise location.

Figure 3: Velocity profiles for three different streamwise locations.

The results show that all three turbulence models predict similar velocity profiles, and they all capture the general behavior and development of the mixing layer well. Small differences can be observed in the sharpness and thickness of the velocity gradient, but overall, each model provides acceptable agreement with theoretical expectations and experimental measurements. This suggests that, for the prediction of mean velocity in a turbulent mixing layer, the choice between $k - \varepsilon$, $k - \omega$, and SST $k - \omega$ models is not critical, and all models are capable of delivering reasonable accuracy.

Reynolds stress components at different streamwise locations

Reynolds stress profiles were extracted at $x = 200$ mm and $x = 650$ mm to analyze the development of turbulent fluctuations within the mixing layer.



(a) Reynolds stress $\overline{u'u'}$ component distribution at $x = 200$ mm streamwise location.

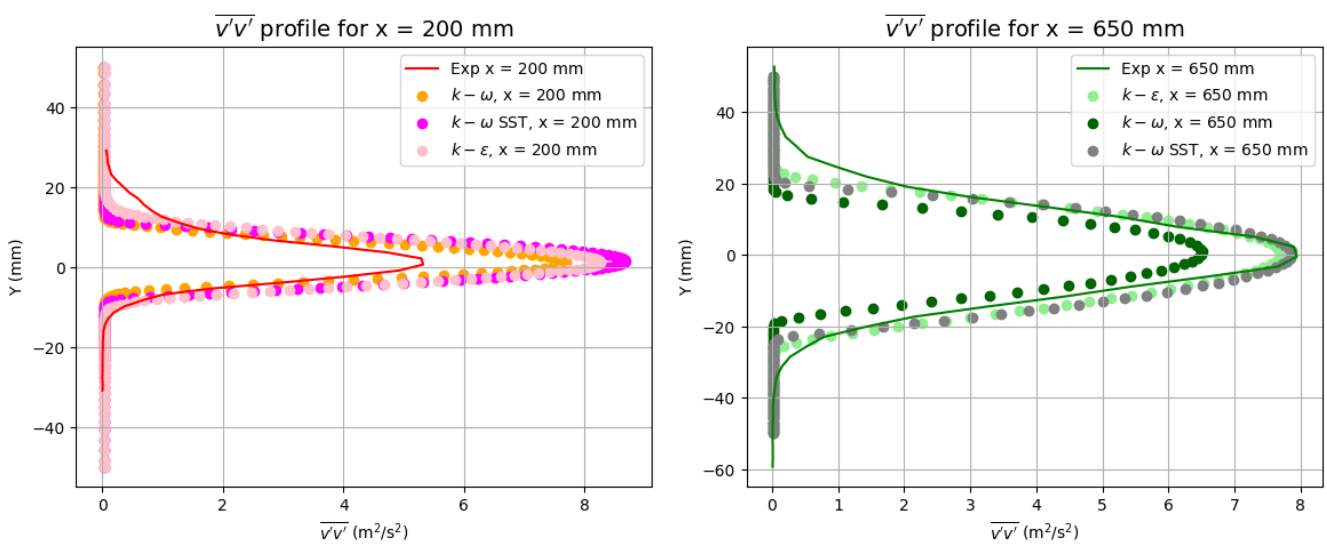
(b) Reynolds stress $\overline{u'u'}$ component distribution at $x = 650$ mm streamwise location.

Figure 4: Reynolds stress $\overline{u'u'}$ component distribution at two different streamwise locations.

At x = 200 mm, Fig. 4a, the $\overline{u'u'}$ distribution shows a sharp peak centered within the mixing layer, indicating intense turbulence production where the velocity gradient is largest. The peak is relatively high and the distribution remains narrow, reflecting the strong shear-driven turbulence near the upstream region. The predicted distributions from all three turbulence models show good agreement with the experimental data. Among the models, the SST $k-\omega$ model provides the closest match to the experimental peak value, while the standard $k-\varepsilon$ and $k-\omega$ models slightly underpredict the peak. However, the discrepancies are small, and overall all models perform reasonably well at this location.

At x = 650 mm, Fig. 4b, the mixing layer has broadened, and the $\overline{u'u'}$ distribution becomes wider and flatter. The peak value increased compared to x = 200 mm. This reflects the continued growth of turbulence as the mixing layer evolves downstream, with larger turbulent structures developing and increased turbulent kinetic energy. These observations are consistent with the expected behavior of a turbulent shear layer, where turbulence production initially dominates and is followed by gradual diffusion and dissipation of turbulent kinetic energy further downstream. All three turbulence models underpredict the Reynolds stress levels compared to the experimental measurements. The SST $k-\omega$ and $k-\varepsilon$ models show very similar behavior, with their profiles closely overlapping each other. The standard $k-\omega$ model underpredicts the peak even more significantly, resulting in the largest discrepancy among the three. This suggests that while the models are able to reasonably capture the early development of turbulence, they tend to underestimate the strength of turbulence fluctuations farther downstream, possibly due to limitations in modeling the growth of large-scale turbulent structures in a mixing layer.

The turbulence models perform well at x = 200 mm because the flow is still in the early development phase, characterized by strong shear-driven turbulence production, relatively isotropic turbulence structures, and a thinner mixing layer. These conditions align closely with the assumptions behind eddy-viscosity-based RANS models, leading to good agreement with experimental Reynolds stress measurements.



(a) Reynolds stress $\overline{v'v'}$ component distribution at $x = 200$ mm streamwise location.

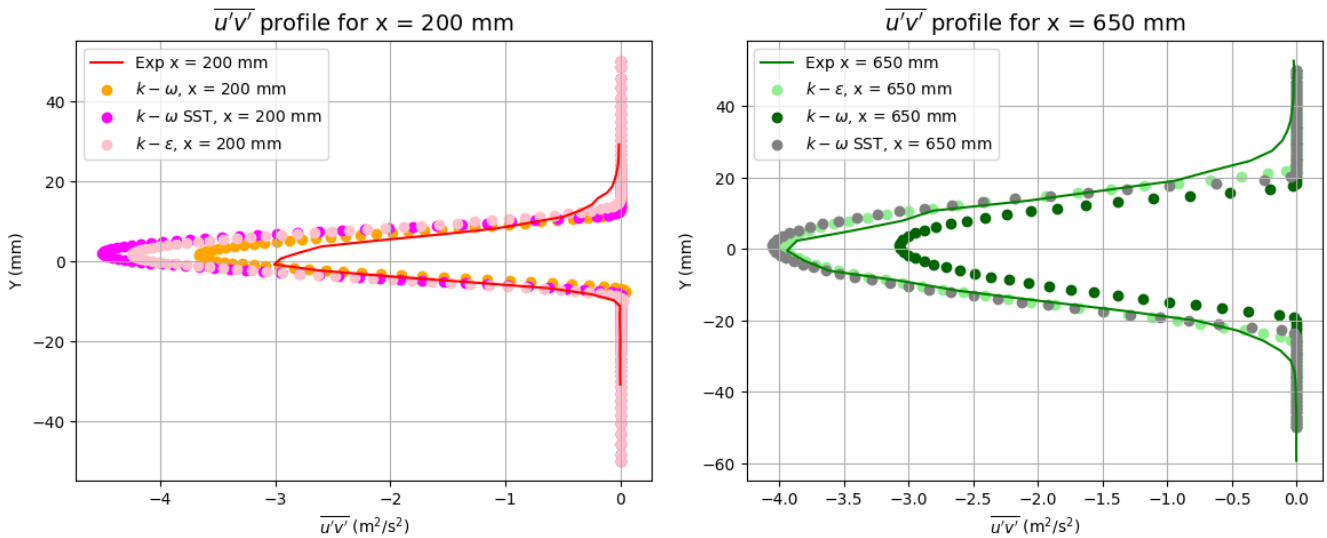
(b) Reynolds stress $\overline{v'v'}$ component distribution at $x = 650$ mm streamwise location.

Figure 5: Reynolds stress $\overline{v'v'}$ component distribution at two different streamwise locations.

The Reynolds stress component $\overline{v'v'}$ represents the fluctuating velocity correlation in the vertical direction. At $x =$

200 mm, Fig. 5a, the results show a distinct peak in the Reynolds stress profile near the shear layer's center, which is typical for turbulent flows where high velocity gradients exist. The peak is relatively high and the distribution remains narrow, reflecting the strong shear-driven turbulence near the upstream region. But the peak is smaller comparing to the $\overline{u'u'}$ distribution of Reynolds stress, indicating weaker turbulence development in that direction. Even though, in this case, all three turbulence models don't show huge discrepancies between themselves, they tend to over-predict the Reynolds stress.

At $x = 650$ mm, Fig. 5b,the mixing layer has broadened, and the $\overline{v'v'}$ distribution becomes wider and flatter. The peak value increased compared to x = 200 mm. This reflects the continued growth of turbulence as the mixing layer evolves downstream, with larger turbulent structures developing and increased turbulent kinetic energy. But compared to the $\overline{u'u'}$ distribution the peak is smaller and it can be attributed to smaller gradient, i.e. weaker turbulence development in y direction. In this case, $k-\varepsilon$ and $k-\omega$ SST models accurately match the experimental values, while the $k-\omega$ underestimates the distribution. This can be atributed to that the $k-\omega$ model uses the turbulence frequency ω and is designed to resolve the near-wall region better by capturing the small turbulent structures. However, in a growing mixing layer, the larger turbulent structures that develop in the far-field are not as well captured by the $k-\omega$ model. The SST model, which is a hybrid of both the $k-\omega$ and $k-\varepsilon$ models, improves this by transitioning to the $k-\varepsilon$ model farther from the wall, where the larger turbulent eddies are more significant, thus providing a better match with the experimental data.



(a) Reynolds stress $\overline{u'v'}$ component distribution at $x = 200$ mm streamwise location.

(b) Reynolds stress $\overline{u'v'}$ component distribution at $x = 650$ mm streamwise location.

Figure 6: Reynolds stress $\overline{u'v'}$ component distribution at two different streamwise locations.

The distribution of $\overline{u'v'}$ represents the Reynolds stress associated with the covariance between the velocity components in the $x-$ and y-directions, often referred to as the shear stress in turbulence. In the context of a mixing layer, the behavior of this component is influenced by the shear in the flow and the interactions between the turbulent structures in the x and y directions.

At $x = 200$ mm, the $\overline{u'v'}$ component exhibits a sharp gradient across the mixing layer, with negative values indicating the momentum transfer between the turbulent structures in the horizontal and vertical directions. This negative correlation is typical for turbulent shear layers, where fluctuations in the u-direction (horizontal velocity) are an-

ticorrelated with those in the v-direction (vertical velocity). The negative values suggest that turbulent eddies are transferring momentum from one stream to the other, leading to shear stress across the layer. The profile shows a well-defined region of negative shear stress near the center of the mixing layer, where the turbulence is most intense, and gradually becomes less negative towards the outer regions of the layer. All three turbulence models overpredict the values of Reynolds shear stress because they tend to overestimate the turbulence intensity in the initial stages of the mixing layer. This is often due to the models' assumption of isotropic turbulence or the inability to fully capture the complexity of the shear layer's development, particularly in the early stage where turbulence is still growing. The models may overestimate the magnitude of velocity fluctuations and shear stress due to insufficient resolution of the turbulent structures near the initial shear zone. As a result, the predicted $\overline{u'v'}$ is higher than what is observed experimentally.

At $x = 650$ mm, the $\overline{u'v'}$ component becomes more spread out, reflecting the further evolution of the mixing layer and the growth of turbulent structures. The distribution at this downstream location is wider and less steep, as the turbulence continues to develop and the mixing layer grows. Despite the broader profile, the negative values of $\overline{u'v'}$ persist, indicating that the turbulent mixing between the two streams continues to be dominated by shear stress interactions. The magnitude of the negative $\overline{u'v'}$ is slightly increased compared to $x = 200$mm. The profile at this location indicates that the turbulence is spreading across a larger area, leading to a reduction in the intensity of the shear stress in the $\overline{u'v'}$-direction. SST $k - \omega$ and $k - \epsilon$ models match the experimental results accurately because they better capture the evolution of the turbulence in the downstream mixing layer, including larger eddies and reduced wall effects. However, the $k - \omega$ underestimates the shear stress because it is more sensitive to near-wall regions and struggles to accurately predict the turbulence in the wider, more developed flow, leading to a weaker prediction of the Reynolds shear stress.

Conclusion

The simulation of the 2D turbulent mixing layer provided valuable insights into the development of turbulence and the performance of different turbulence models. The results show that the velocity magnitude distribution exhibits clear freestream regions and a developing mixing layer that grows in thickness downstream. The velocity profiles extracted at different streamwise locations confirm the gradual transition of velocities and the typical behavior of a turbulent shear layer.

The turbulence models, including the standard $k-\varepsilon$, $k-\omega$, and SST $k-\omega$ models, all predicted the overall evolution of the mixing layer well. The models showed good agreement with experimental data for mean velocity profiles at various streamwise locations. While minor differences were observed, particularly in the sharpness and thickness of the velocity gradient, all models provided reasonable predictions for the velocity profile in the turbulent region.

The Reynolds stress components were also analyzed at different streamwise locations to evaluate the turbulence intensity and shear stress. The $k - \varepsilon$ and SST $k - \omega$ models provided the best agreement with experimental data, capturing the turbulent behavior effectively in the downstream regions. In contrast, the standard $k - \omega$ model tended to underestimate the turbulence intensity, particularly in the later stages of the mixing layer. This can be attributed to its inherent focus on near-wall turbulence structures, which are less significant in the far-field of the mixing layer. But for $x = 200$ mm, all models tend to overpredict the $\overline{u'v'}$ and $\overline{v'v'}$ Reynolds stresses. Exception is $\overline{u'u'}$ at $x = 200$ mm and $x = 650$ mm, where the turbulence presence is the strongest. This problem arises due to the models' isotropic treatment of velocity gradients.

The Reynolds shear stress $\overline{u'v'}$ was overpredicted by all three models, especially in the early development stages of the mixing layer, where turbulence intensity is still growing. However, as the mixing layer developed downstream, the models provided better matches to the experimental data, especially the SST $k - \omega$ and $k - \varepsilon$ models.
In conclusion, all three turbulence models provided valuable insights into the turbulent mixing layer, with the

SST $k-\omega$ model offering the most accurate predictions for velocity profiles and Reynolds stress components in the developed mixing layer. The results demonstrate that while the choice of turbulence model is important, the performance differences between the models are relatively small for capturing the overall behavior of the mixing layer, especially when comparing $k-\varepsilon$, $k-\omega$, and SST models for turbulent flows in shear layers.

References

[1] Garem J. H. Delville J., Bellin S. and Bonnet J. P. 2dml: 2d mixing layer validation case. `https://turbmodels.larc.nasa.gov/delvilleshear_val.html`, 1989.

[2] Inc. ANSYS. *Standard k-epsilon Model*, 2025.

[3] Inc. ANSYS. *Standard k-omega Model*, 2025.

[4] Inc. ANSYS. *SST k-omega Model*, 2025.